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Development of a Smart, Safer and Cost-Effective Technology for Enhanced Fugitive Emission Monitoring

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Abstract

Fugitive emissions, particularly methane and Volatile Organic Compounds (VOCs), present critical environmental and economic challenges, contributing to climate change and possibly incurring regulatory penalties. A Leak Detection and Repair (LDAR) program is crucial for minimizing fugitive emissions and ensuring the safety and efficiency of these facilities. Chiyoda Corporation is a worldleading engineering company and Engineering, Procurement, and Construction (EPC) contractor for large-scale projects across the globe. In these projects, there are thousands of fugitive emission sources, and using conventional LDAR programs that rely on periodic manual inspections of individual components (e.g., EPA Method 21) has disadvantages, such as being time-consuming, incurring high labor costs, delayed leak identification, and the introduction of potential human data-entry errors.

We are developing a smart LDAR technology that achieves real-time, high-accuracy leak detection through the integration of gas leak detection devices, 3D digital twin software, and robotic platforms; optimizes operational costs and reduces downtime through proactive maintenance strategies. It also ensures compliance with evolving LDAR regulations, including US-EPA, EU methane strategy, OGMP 2.0 and API 624 requirements, ensuring effective monitoring and management of methane and other fugitive emissions. The proposed technology combines three core components: (1) gas leak detection sensors—such as Optical Gas Imaging (OGI) cameras and Tunable Diode Laser Absorption Spectroscopy (TDLAS) sensors—to detect, quantify, and visually identify gas leaks; (2) 3D digital twin software to develop an LDAR database by incorporating fugitive emission data (component type, location, concentration), enabling prioritized repair scheduling and optimized inspection routing; and (3) robotic platforms for continuous, autonomous monitoring over larger areas, collecting more data points than manual inspections and accessing challenging or hazardous zones for more accurate leak- flux quantification.

The first demonstration study of the proposed technology was conducted at Chiyoda Corporation's SPERA Hydrogen demonstration plant in Japan. Results demonstrated the capabilities of the proposed technology for fugitive emission leak detection and quantification through safe and efficient monitoring without requiring proximity to potential emission sources. Future developments aim to integrate an

autonomous robot instead of a manually controlled one, enabling continuous, high-precision monitoring, especially in hazardous or hard-to-access industrial environments. Our work will further enhance LDAR efficiency, improves safety, ensures regulatory compliance, and ultimately contributing to environmental sustainability and economic benefits.

Introduction

Fugitive emissions refer to the unintended release of gases or vapors from pressurized equipment, pipelines, and other industrial sources. These emissions, particularly methane and Volatile Organic Compounds (VOCs), pose significant environmental and economic challenges, contributing to climate change and potentially resulting in regulatory penalties. Monitoring and controlling fugitive emissions are essential for environmental protection and regulatory compliance (International Energy Agency, 2020). Leak Detection and Repair (LDAR) technology plays a crucial role in managing fugitive emissions, supporting compliance with environmental regulations and contributing to sustainability efforts. It is a vital component of modern industrial practices.

Chiyoda Corporation is a world-leading engineering company and EPC contractor traditionally delivered large-scale LNG and refinery projects, in addition to recently added renewable energy projects. In these projects, there are thousands of fugitive emission sources, such as flanges, valves, compressors and pumps. Using conventional LDAR programs, which rely on periodic manual and direct inspections of each individual component (e.g., EPA Method 21), face several disadvantages such as time-consuming, costly, delaying leak identification, introducing potential human data-entry errors and pose safety risks due to manual inspections.

As shown in the table 1, in comparison with conventional LDAR, Smart LDAR technology represents an evolution of conventional practices, leveraging advanced technologies and data analytics to enhance the efficiency and effectiveness of fugitive emissions monitoring. It employs state-of-the-art technologies, such as OGI cameras and TDLAS sensors, which are particularly effective for detecting methane and VOC emissions while significantly improving safety and reducing monitoring time and costs.

Table 1—shows a comparison between conventional LDAR and Smart LDAR

Criteria	Conventional LDAR	Smart LDAR
Operational Costs	- Lower initial costs for traditional methods - Higher ongoing costs due to manual inspections and labor-intensive processes	- Initial investment in advanced technology (sensors, software) - Lower ongoing costs due to automation and real-time monitoring - Potential for reduced workforce requirements
Maintenance Savings	- Reactive maintenance approach, leading to higher costs when issues arise - Longer repair times which can increase operational disruptions	- Predictive maintenance capabilities leading to reduced maintenance downtime - Faster detection and repair of leaks, minimizing product loss - Optimized resource allocation due to data analytics
Regulatory Penalty Avoidance	- Higher risk of non-compliance due to manual inspections - Potential for fines and penalties if leaks are missed or reporting is inaccurate - Increased burden for compliance documentation and audits	- Enhanced compliance through real-time monitoring and reporting - Reduced risk of leaks going undetected, lowering the chances of fines - Ability to provide data for regulatory audits easily
Overall ROI	- Short-term cost savings but potential long-term financial risks due to penalties and inefficiencies - Possible reputational damage from non-compliance	- Long-term cost savings from reduced leaks and penalties - Improved efficiency and productivity - Enhanced reputation as a proactive and environmentally responsible organization

The integration of 3D digital twin in Smart LDAR technology has emerged as a transformative approach for monitoring fugitive emissions. By utilizing this technology, industries can enhance their understanding of emission sources and patterns, leading to more effective leak detection and management strategies. Research has demonstrated that 3D digital twins can significantly improve the accuracy of emissions monitoring. For instance, implementation of a data-driven digital twin method for leak detection in natural gas pipelines has shown promising results in enhancing the accuracy and efficiency of emissions monitoring (Liang et al., 2023)

Guido et al. (2024) showcased the capabilities of TDLAS sensors installed on robots for methane detection and quantification in controlled gas flow rates facility. Ilonze et al. (2024) tested the accuracy and repeatability of FLIR GF320 OGI in combination with FLIRQL320 QOGI. The results showed that, approximately 75% of measurements fell within a quantification error.

Eric et al. (2019) investigated the field deployment of a portable optical absorption spectrometer for the localization and quantification of fugitive methane (CH₄) emissions, demonstrating the effectiveness of TDLAS and customized VOC sensors in identifying leak sources. Martinez et al. (2020) presented the development, integration, and testing of methane sensors for deployment on small unmanned aerial systems (sUAS), illustrating the capability of sUAS for emissions monitoring and quantification from natural gas infrastructure, thereby providing a more efficient method for detecting leaks and ensuring regulatory environmental compliance while enhancing safety protocols.

Junjie et al. (2024) explored a detection approach for natural gas leakage from offshore platforms by integrating OGI cameras with unsupervised deep probability learning, achieving higher detection accuracy and real-time capabilities with an impressive inference time of just 4 seconds per frame.

Georg. et al. (2017), introduced an autonomously navigating robotic system equipped with diverse sensors was designed for comprehensive plant inspection. Their study demonstrating the capability to detect methane leaks as small as 40 mL/min and achieving an average precision of 76.62% in 2D object detection.

In this paper, we present Smart LDAR technology for fugitive emission detection and quantification using OGI cameras, integrated with 3D digital twin software and robotics. This integration of digital twin technology allows for the creation of virtual representations of physical assets, enabling real-time monitoring and analysis of emissions data. This not only enhances the accuracy of leak detection but also facilitates predictive maintenance, reducing downtime and operational costs. Furthermore, the incorporation of robotics into Smart LDAR technology allows for continuous, autonomous, and data-driven leak management in hard-to-access or hazardous locations, ensuring comprehensive monitoring without compromising safety.

Methodology

The proposed technology employs gas leak detection sensors to detect, quantify, and visually identify gas leaks, integrating with digital twin software to develop an LDAR database and robotic platforms. This combination allows for monitoring over larger areas, collecting more data points than manual inspections, and accessing challenging or hazardous zones for more accurate leak-flux quantification. Such integration facilitates proactive maintenance strategies and leads to cost savings over time by minimizing downtime, reducing labor costs, and ensuring compliance with environmental regulations.

Gas leak detection Sensor

Advanced gas leak detection sensors, such as OGI cameras, utilized for both gas leak detection and quantification. Infrared thermal imaging cameras compare the difference in background infrared radiation and the radiation emitted by the hydrocarbon gas being detected. The technology recognizes the absorbing and emission properties of radiation in the infrared spectrum of the detected gas and the camera produces a thermal image. Conventional infrared thermal imaging cameras also visualize non-gaseous radiation

sources, such as plant facilities or personnel, and produce their thermal image. Modern advanced cameras are designed to only visualize and create thermal images of dispersing gas clouds in motion.

These technologies allow for the capture of real-time gas leak enabling operators to quickly and accurately identify leaks. When cameras are used, multiple sources of leakage can be monitored within the plant area based on the capture of the gas leaks information, addressing the conventional LDAR disadvantage in which each fugitive emission component need to be monitored individually through plant walk. This is expected not only to increase the efficiency of maintenance work, but also to reduce the number of leaks missed. Furthermore, the visual data collected can be integrated into LDAR databases, facilitating improved analysis and reporting of emissions trends.

OGI cameras can be integrated with data analytics platforms and Internet of Things (IoT) technologies. This integration allows for real-time data collection, analysis, and reporting, enhancing decision-making processes and improving overall monitoring effectiveness.

Market survey for best OGI camera Provider. The purpose of this survey was to evaluate and compare the advantages of various OGI camera providers. We sent the survey to 25 providers to gather detailed information on their camera specifications. The results revealed that some providers excelled in thermal sensitivity and resolution, earning high marks for real-time leak detection and exceptional customer support, making them preferred choices for performance-oriented organizations. Others were recognized for their cost-effectiveness, appealing to budget-conscious users, although feedback indicated a need for more intuitive interfaces. Additionally, certain providers were praised for their OGI camera capabilities, such as quantification and cloud storage availability, which enable seamless data storage and transfer. Based on the insights gathered, we selected the optimum technology provider that aligns with our needs, ensuring access to the most effective and reliable OGI camera solutions for our operations.

3D Digital Twin Software

The 3D Digital Twin replicates real-world equipment used in plant maintenance within a three-dimensional (3D) cyber space. Serving as a gateway to various plant data, the 3D Digital Twin enables seamless access to abnormal detection information, diagnostic data, and equipment details visualized through IoT solutions and robots by incorporating these insights with 3D spatial information. This technology facilitates data-driven plant operations, enhancing efficiency and decision-making processes.

The V-Plant 3D digital twin software enables the creation of a digital replica of a real-world plant is employed to develop the LDAR database, which incorporates information such as the types and number of fugitive emission components, gas leak concentration measurements, monitoring dates, repair date and repair status. The data visualization tools within the digital twin environment enhance communication among operations and maintenance teams, facilitating easy identification of leak components and allows maintenance teams to effectively prioritize repairs for high-leak components.

The V-Plant 3D digital twin can be integrated with various technologies, such as OGI cameras and IoT sensors. This creates a comprehensive monitoring system that enhances the effectiveness of LDAR program. Automatic color coding on 3D models can be implemented based on the status of LDAR components. The insights generated from the immersive experience in the 3D digital twin can support decision-making processes with data driven, helping operators prioritize maintenance activities and allocate resources effectively maintenance.

Robotic Platform

The integration of robotics into Smart LDAR technology represents a truly significant and transformative advancement in the field of environmental monitoring and compliance practices within various industries. By leveraging advanced autonomous robots equipped with sensors and technologies, it is possible to effectively identify, locate, and quantify leaks in real-time even in challenging or hazardous environments that would otherwise pose considerable risks to human workers. Additionally, robots can operate

continuously, providing real-time monitoring and data collection and can be connected to Internet of Things (IoT) platforms, allowing for seamless analysis of vast amounts of data, thereby improving the accuracy and reliability of leak detection processes and facilitating prompt and efficient repairs. This connectivity extended to the 3D digital twin systems and enhances decision-making processes and enables proactive maintenance.

The demonstration Results and Discussion

To assess the reliability of our technology for detecting and quantifying fugitive emissions, a demonstration study was conducted at Chiyoda corporation's SPERA hydrogen demonstration plant in Japan which represents a pioneering effort in the advancement of hydrogen technology, showcasing innovative approaches to sustainable energy production.

As a key facility in Chiyoda Corporation's commitment to developing a hydrogen-based economy, the plant serves as a critical platform for research and development in hydrogen storage, transportation, and utilization. By leveraging cutting-edge technologies and methodologies, the Koyasu plant not only demonstrates the feasibility of large-scale hydrogen production but also addresses the pressing need for environmentally friendly energy solutions. This plant has a hydrogen generation capacity of 50 Nm³/h and utilizes the LOHC (Liquid Organic Hydrogen Carrier) technology, which was developed for the safe and efficient transport of hydrogen. This involves using organic compounds that can absorb hydrogen, allowing for easier storage and transportation compared to that for gaseous and liquified hydrogen.

The demonstration was carried out at both of the pilot plant and the toluene storage tank sampling points to detect gas leaks during sampling time to ensure the presence of VOCs which is crucial for demonstrating the OGI detection capabilities.

Methodology Set up

Figure 1 illustrates the methodology employed in the demonstration study to detect fugitive emission leaks. In this setup, the OGI camera (Konica Minolta GMP02) is securely mounted on a remotely controlled robot using a versatile payload mounting bracket, which offers a high degree of flexibility. This design allows for precise adjustments of the sensor's orientation—specifically roll, pitch, and yaw—enabling optimal positioning according to the specific inspection locations. This flexibility is crucial for ensuring thorough and effective monitoring of potential leak sources, thereby enhancing the overall accuracy of the detection process.

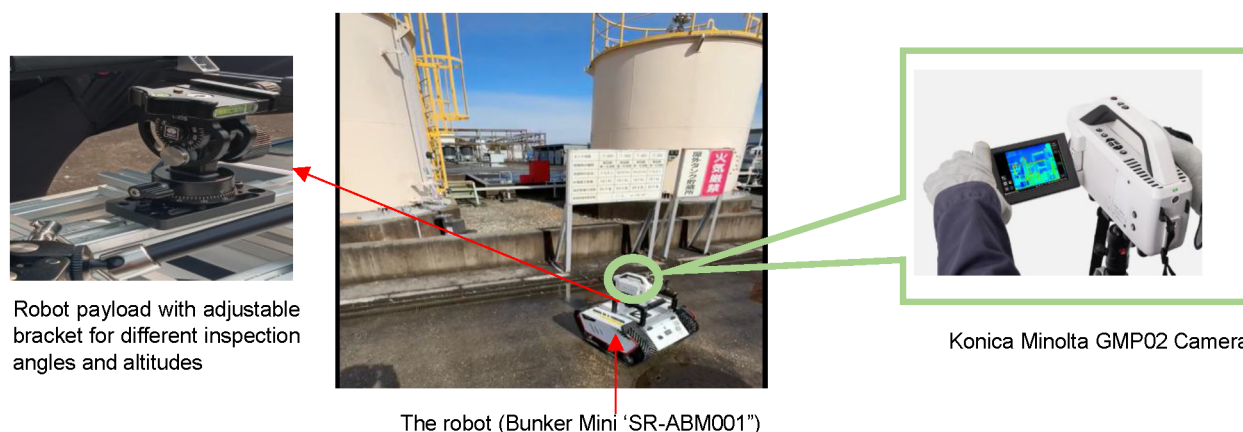


Figure 1—Methodology Setup shows an OGI Camera installed on the robotic platform

The demonstration results

We demonstrated different modes of the OGI camera for the gas leak detection, particularly the infrared detection mode which can provide consistent and reliable monitoring of gas emissions and particularly effective for detecting VOCs and other gases that absorb infrared light. The other, high sensitivity detection mode ensures that even the smallest leaks are identified.

As illustrated in Figure 2, gas leaks can be effectively identified through both the infrared gas detection mode and the high-sensitivity detection mode of the OGI camera. within the detection time of less than 3 seconds. The demonstration was conducted in an open area, which allowed for the simulation of a real environmental conditions. There were fluctuations in measurements due to weather factors such as wind; however, the gas leak was successfully identified and quantified. The measured leak rate ranged from 3 to 9.1 mg/s at a distance of 2 to 3 meters from the leak source. The detection sensitivity (in case of methane) 10% LEL • m (Lower Explosive Limit at a distance of 1 meter).

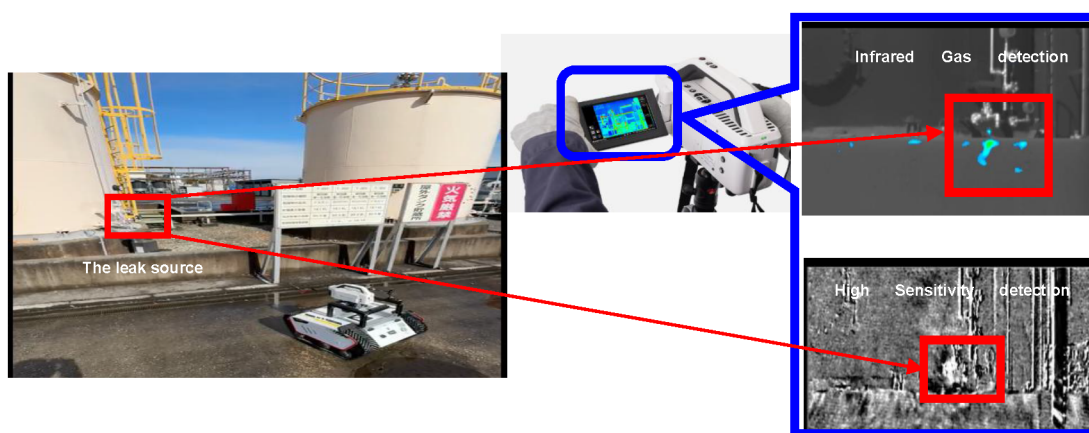


Figure 2—Gas leak detection at toluene storage tank sampling point

Furthermore, we demonstrated the impact of excessive vibrations caused by robot movement on the stability and positioning of the OGI camera, which leads to inaccuracies in gas detection and visualization. To mitigate this issue, it was crucial to stop the robot movement at the time of data collection. This eliminates the effects of vibration, thereby enhancing the precision of OGI detection and allowing for more effective gas emissions monitoring.

Konica Minolta' OGI camera has been tested and deemed as meeting the requirements of US-EPA NSPS OOOOa requirements and was certified as an Approved Instrument Monitoring Method (AIMM) by the Colorado Department of Public Health and Environment (CDPHE), ensuring the high accuracy of the gas leak detection and quantification.

Future Development

This technology is still under development, and we remain committed to utilizing best practices to enhance its capabilities through the integration of advanced technologies, such as 3D digital twins and autonomous robots. These innovations not only improve operational efficiency but also significantly reduce environmental impact, paving the way for a more sustainable future.

LDAR' 3D digital twin database

The LDAR database will be developed using V-Plant 3D digital twin software, which will enable visual representation of leaks for effective repair prioritization through color coding based on gas leak concentration. This visualization empowers maintenance teams to swiftly identify high-leak components,

allowing for optimized repair scheduling, efficient inspection routing, and ultimately, reduced maintenance downtime and costs.

Autonomous robot

Autonomous robotic systems can be programmed for routine and scheduled inspections, significantly enhancing operational efficiency and drastically reducing the time required for traditional manual checks. The deployment of autonomous robots enables continuous, high-precision monitoring, especially in hazardous or hard-to-access industrial environments. Moving forward, we will incorporate advanced gas detection sensors and OGI technologies providing highly accurate measurements and quantification of gas leaks

This integration not only assists in minimizing greenhouse gas emissions and ensuring strict adherence to regulatory standards but also contributes to substantial cost savings for companies by optimizing maintenance schedules and reducing the need for labor-intensive and time-consuming processes. Ultimately, the utilization of advanced robotics in LDAR technology marks a transformative and pivotal step to improve plant operation safety, ensures regulatory compliance, and ultimately contributes to environmental sustainability and economic benefits.

Scaling the technology from demonstration to full industrial deployment

The below steps will be carried out for scaling our technology from demonstration phase to full industrial deployment:

Pilot Program in partnership with potential customers.

- Select the sensors and autonomous robots to be implemented and prepare the 3D-digital twin data of the facility to be tested.
- Test the integration of 3D digital twins and autonomous robots occupied with advanced gas detector in a controlled environment at the potential customer site.
- Gather extensive data on leak occurrences and repair efficiency during the pilot program.
- Validate the accuracy of gas leak quantification by comparing the results obtained from the autonomous robotic detection sensor with those from portable leak detectors.

Broader Implementation Strategy.

- Refine the technology based on data analysis and validation outcomes to optimize operational processes.
- Develop a comprehensive deployment strategy for scaling the technology across designated industrial sites.

Full Industrial Deployment. As a final step, we will be able to seamlessly deploy our enhanced LDAR technology across all designated industrial sites.

Conclusions

The results of the first demonstration study of the proposed technology, conducted at Chiyoda Corporation's SPERA Hydrogen demonstration plant in Japan, showcased the remarkable capabilities of this technology for detecting fugitive emissions. This study highlighted the ability to perform safe and efficient monitoring without necessitating proximity to potential emission sources, thereby enhancing overall safety for personnel involved.

Ongoing research and development activities are actively underway to refine and expand this innovative technology as part of Chiyoda Corporation's plant operation and maintenance service, known as plantOS.

Future developments aim to incorporate an autonomous robot in place of a manually controlled one, enabling continuous, high-precision monitoring, particularly in hazardous or hard-to-access industrial environments. Through these advancements, our work is set to significantly enhance LDAR efficiency, improve safety standards, ensure compliance with regulatory requirements, and ultimately contribute to environmental sustainability and tangible economic benefits.

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